

Device Facts

Apis

K182725

Applicant	Sweetbio, Inc.
Decision Date	May 31, 2019
Product Code	FRO
Decision	SESE
Submission Type	Traditional
Review Panel	SU
Device Class	Class U

ATTRIBUTES

- x AI / ML

x SaMD

x HFE Study

x PCCP

x RWE

+ Therapeutic

x Pediatric

x 3rd Party
- x Expedited

INDICATIONS FOR USE

The intended use of Apis in the management of wounds is to cover and protect the wound, absorb wound exudate, and provide and maintain a moist wound environment.

Device Narrative

DEVICE STORY

Apis is a sterile, single-use, conformable, biodegradable, and absorbable wound dressing. It consists of a crosslinked porcine-derived gelatin (collagen derivative) matrix reinforced with hydroxyapatite and Manuka honey. The device is applied to wounds to provide a moist healing environment, absorb exudate, and protect against shear, friction, and pressure. It is intended for use by clinicians in a professional setting. The device degrades in the wound bed via natural enzymatic and hydrolytic processes. It functions as both a protective cushion and a dressing to facilitate wound management.

CLINICAL EVIDENCE

No clinical data. Evidence consists of non-clinical bench testing including fluid uptake, conformability, porosity, mechanical testing, endotoxin levels, and chemical composition. Biocompatibility testing included cytotoxicity, sensitization, irritation, acute systemic toxicity, material-mediated pyrogenicity, subchronic toxicity, and implantation. Chronic toxicity, genotoxicity, and carcinogenicity were addressed via chemical characterization and toxicological risk assessment.

TECHNOLOGICAL CHARACTERISTICS

Sterile, single-use, biodegradable sheet. Materials: Manuka honey, crosslinked porcine-derived gelatin, and hydroxyapatite. Dimensions: 16x16 mm, 25x25 mm, 50x50 mm. Function: Absorbable wound dressing. Sterilization: Gamma radiation.

Lineage

PREDICATE DEVICES

Elasto-Gel Manuka Honey Wound Dressing (K083334)

REFERENCE DEVICES

Collagen Wound Dressing (K112580)

Regulatory Classification

SU	FRO
Device Class	Class U
Review Panel	SU



May 31, 2019

SweetBio, Inc
% Carolyn Guthrie
Director of Quality/Regulatory
Kapstone Medical, LLC
PO Box 969
Leicester, North Carolina 28748

Re: K182725
Trade/Device Name: Apis
Regulatory Class: Unclassified
Product Code: FRO
Dated: April 26, 2019
Received: April 29, 2019

Dear Carolyn Guthrie:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database located at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part

801); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803) for devices or postmarketing safety reporting (21 CFR 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR 1000-1050.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR Part 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

For David Krause
Acting Division Director
DHT4B: Division of Infection Control
and Plastic Surgery Devices
OHT4: Office of Surgical
and Infection Control Devices
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K182725

Device Name

Apis

Indications for Use (Describe)

Apis is indicated for use in management of wounds, including

- full and partial thickness wounds
- pressure ulcers (stages I-IV)
- venous stasis ulcers
- diabetic ulcers
- abrasion
- surface wounds
- traumatic wounds (healing by secondary intention)
- donor site wounds
- surgical wounds

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

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510(k) Summary

In accordance with Title 21 of the Code of Federal Regulations, Part 807, and in particular 21 CFR 807.92, the following summary of information is provided.

1. Date of Preparation:

28 May 2019

2. Applicant

SweetBio, Inc
20 Dudley Street, Suite 900
Memphis, TN 38103

3. Official Correspondent

Kapstone Medical LLC
5960 Fairview Rd, #400
Charlotte NC 28210

Contact Person:

Carolyn Guthrie
Email: cguthrie@kapstonemedical.com

4. Device Name

Trade Name: Apis
Common/Usual Name: Dressing, Wound, Drug
Classification: Unclassified
Regulation Number: Unclassified
Product Code: FRO
Panel: General and Plastic Surgery

5. Predicate Devices

Apis is substantially equivalent to the predicate devices shown in Table 1-1.

Predicate	510(k) Number	Device	Manufacturer
Primary	K083334	Elasto-Gel Manuka Honey Wound Dressing	Southwest Technologies, Inc
Reference	K112580	Collagen Wound Dressing	Dalim Tissen Co., Ltd

Table 1-1: Predicate devices

6. Device Description

Apis, for the management of wounds, is a sterile, single-use device. Apis is a conformable solid sheet that is biodegradable and absorbable. Apis, in its final form, is comprised of a highly purified collagen derivative (gelatin) from porcine skin, Manuka honey, and hydroxyapatite (HAp).

The intended use of Apis in the management of wounds is to cover and protect the wound, absorb wound exudate, and provide and maintain a moist wound environment. The structural reinforcement of Apis is achieved through the incorporation of hydroxyapatite into the crosslinked collagen derivative structure. The absorption and moisture-retaining properties are based on the manufacturing process and incorporation of honey into the crosslinked structure.

The device is biodegradable and absorbs into the surrounding tissue. The degradation occurs through natural processes such as enzymatic breakdown, hydrolysis and other chemical/biological reactions depending on wound type and level of exudation.

7. Indications for Use

Apis is indicated for use in management of wounds, including:

- Full and partial thickness wounds
- pressure ulcers (stages I -IV)
- venous stasis ulcers
- diabetic ulcers
- abrasions
- surface wounds
- traumatic wounds (healing by secondary intention)
- donor site wounds
- surgical wounds

8. Technological Characteristics

The subject device was shown to be substantially equivalent and have the same intended use and similar technological characteristics to its predicate devices through comparison in areas including design, material composition, function and sterilization method.

Features	Apis	Elasto-Gel™ Manuka Honey Wound Dressing (sheet)	Substantial Equivalence
	Subject Device	Primary Predicate Device	
Indications for Use			
Intended Use	Management of wounds	Management of wounds	Equivalent
Type of Use	Prescription Use	Prescription Use	Equivalent
Indications	• full and partial thickness wounds	• full and partial thickness wounds	Equivalent

Features	Apis	Elasto-Gel™ Manuka Honey Wound Dressing (sheet)	Substantial Equivalence
	Subject Device	Primary Predicate Device	
	• pressure ulcers (stages I-IV) • venous stasis ulcers • diabetic ulcers • abrasions • surface wounds • traumatic wounds (healing by secondary intention) • donor sites wounds • surgical wounds	• pressure ulcers (stages I-IV) • venous stasis ulcers • diabetic ulcers • abrasions • surface wounds • traumatic wounds (healing by secondary intention) • donor sites wounds • surgical wounds	
Single Use	Yes	Yes	Equivalent
Sizes	16 x 16 mm 25 x 25 mm (not for use on children) 50 x 50 mm (not for use on children)	Supplied in many sizes, for example 2x3", 4x4", 6x8" and possibly other additional shapes and sizes.	Comparison to predicate device shows no new questions of safety or effectiveness, and no change to intended use compared to predicate or reference device
Materials			
Materials of Construction	Manuka Honey	Manuka Honey	Equivalent
	Crosslinked highly purified collagen derivative (gelatin) from porcine skin	Insoluble crosslinked polyacrylamide polymer	Comparison to reference device shows no new questions of safety or effectiveness, and no change to intended use compared to predicate or reference device
	Hydroxyapatite	Glycerin	Based on the function of the hydroxyapatite, the intended use of the device is not changed compared to the predicate. Additionally, Apis <i>in vivo</i> animal data shows that there are no new questions of safety or effectiveness
Technological Characteristics			

Features	Apis	Elasto-Gel™ Manuka Honey Wound Dressing (sheet)	Substantial Equivalence
	Subject Device	Primary Predicate Device	
Absorbency	A relatively high capacity for absorption of the wound fluid	A relatively high capacity for absorption of the wound fluid	Equivalent
Absorbable	Fully absorbable	Partially absorbable	Comparison to predicate device shows no new questions of safety or effectiveness, and no change to intended use compared to predicate or reference device
Moist environment	Maintains moist wound environment	Maintains moist wound environment	Equivalent
Wound protection	Protects the wound from shear, friction and pressure and is suitable as a protective padding and cushioning device as well as functioning as a dressing.	Protect the wound from shear, friction and pressure, suitable as a protective padding and cushioning device as well as functioning as a dressing.	Equivalent
Performance			
Sterility	Yes, gamma sterilization	Yes, gamma sterilization	Equivalent

9. Performance Data

Nonclinical testing was performed to demonstrate that the subject device, Apis, is as safe and effective as the legally marketed predicate device, including fluid uptake, conformability, porosity testing, mechanical testing, endotoxin level testing, and chemical composition testing. Sterilization validation, distribution testing and shelf-life testing have been completed.

Biocompatibility testing and literature demonstrated that Apis is safe for its intended use. Cytotoxicity, sensitization, irritation, acute systemic toxicity, material-mediated pyrogenicity, subchronic toxicity, and implantation testing were performed. Chemical characterization and toxicological risk assessment were performed to address the chronic toxicity, genotoxicity, and carcinogenicity endpoints.

10. Conclusions

Based on the indications for use, technological characteristics, and comparison to predicate devices, Apis has been shown to be substantially equivalent to legally marketed predicate devices.